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Storification in History education: A mobile game in and about medieval Amsterdam

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ABSTRACT

A mobile and multimedia game designed for History education was analyzed in terms of how it is designed and how it was applied as a narrative learning environment. In History education, narrative can be argued to be very useful to overcome fragmentation of the knowledge of historical characters and events, by relating these with meaningful connections of temporality and sequence (storification). In the game studied, students explore the history of Amsterdam by walking in the city, experiencing characters, buildings, and events, while using UMTS/GPS phones for communication and exchange of information. The History game was played during one day by 216 students, spread over 10 secondary school classes, in groups of four or five students. All information exchanged during the games was collected, and the game play and introduction of the game was observed by team coaches and researchers. The design of the game as well as the actual gaming process was analyzed with respect to how it evoked three types of storification: receiving (spectator), constructing (director) and participating in (actor) the story. Results show that the game evoked a mixture of these three types of storification. Moreover, these types of storification processes differently affected students' engagement. Participating in the story evoked high activity in the game but less awareness of the whole story, whereas constructing the story triggered awareness of the whole story. Compared to receiving the story, both these types positively affected the engagement of the students being active and motivated during the game.

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The failure to grasp the nature of historical context is often described as an important source of student misunderstanding (c.f., Wineburg, 2001). Students often borrow a context from their contemporary world and view the past through the lens of the present. Therefore, situating a historical phenomenon in a historical context can be considered as a key activity in History education. Barton and Levstik (2004) show that a narrative is particularly relevant for History education, as it is a valuable means to move beyond knowing fragmented facts of historical figures and events. A narrative organizes historical information in a coherent way and, therefore, helps students to remember and recall these events and processes (Falk & Dierking, 2000; Mandler, 1984; Sugiyama, 2001). Moreover, narratives have been argued valuable for learning because they not only incorporate knowledge, but also engage pupils, thereby addressing the emotional side of learning (Aylett, 2006). Games are often mentioned as offering fruitful narrative learning environments (Jenkins, 2004). However, not much is known about the values of games as a narrative learning environment. In the present study, we evaluate how narrative was designed and applied in a mobile History game.

1. Narrative and game-based learning

Narrative has long been recognized as a valuable means for structuring and giving meaning to experiences (Hermans & Hermans-Jansen, 1995). It is nevertheless difficult to find a definition of narrative that is widely accepted. We reference the work of Aylett (2006) for an understanding of narrative. As Aylett stated, "From a perspective centred on learning, one can take a view of story as a process of internal structuring of experience; of sense-making via narrative organization" (p. 6). In line with Aylett, we call this process 'storification'. Storification not only involves structuring, but also simultaneously making sense of experiences. Moreover, storification is a means to organize episodes, actions, and accounts of actions in time and space (Sarbin, 1986). More specifically, Aylett points to the continuous interplay between character and causation in storification (see Fig. 1). Central in storification are questions like "who is doing what and when, with what intentions, and what are the consequences"? As she notes, characters relate to aspects of personality, emotional state and social position. Characters are linked to causation via motives, intentions, plans and actions. Causation denotes issues as temporality and sequence,

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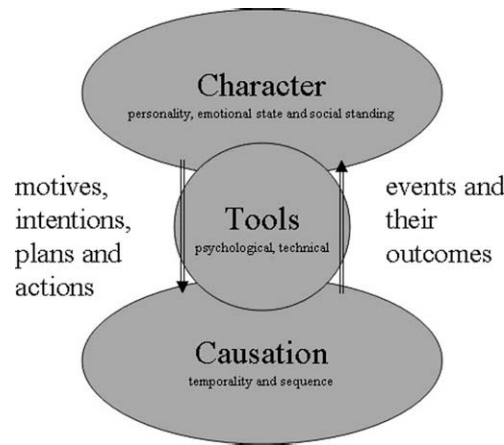


Fig. 1. Narrative organization.

and is linked to characters by events and their outcomes. In line with Aylett, we state that characters and causation are central elements in storification processes. In addition, we argue that it is important to point out tools as mediating the interplay between characters and causation. *Vygotsky (1981)* emphasized that tools, both technical and psychological are crucial elements in human behavior:

“By being included in the process of behaviour, the psychological tool alters the entire flow and structure of mental functions. It does this by determining the structure of a new instrumental act, just as a technical tool alters the process of a natural adaptation by determining the form of labour operations” (p. 137).

In the context of stories, we argue that tools mediate, on the one hand, what characters are triggered to do or capable of doing, and thereby influence their motives, intentions, plans and actions. On the other hand, tools mediate how actions turn out, and influence events and outcomes. To summarize, narrative organization can be analytically described as structuring the interplay between characters, tools and causation. These narrative elements can be pictured as in Fig. 1. Storification entails the process of transforming human experience into story lines by making sense via narrative organization.

Fig. 1 depicts the general nature of storification processes as a narrative organization. In their review *Dettori, Gianetti, Paiva, and Vaz (2006)*, describe three ways in which storification has been used in learning: receiving a story, constructing a story and participating in a story. Though these three different ways of addressing a story are simplifications, they allow us to point out and analyze different roles of the learner. For example, reading and watching are types of receiving a story, as the story is fixed and defined by others, and presented as self-explanatory. In case of receiving a story, students can listen or look at a presented narrative, but not influence it. The student is no part of the story and acts as a spectator. The storification process in these cases entails paying attention and reconstructing the narrative message that is presented. In contrast, when the learning environment entails constructing a story, the student defines the story in the role of the author. In this case, storification entails building the narrative structure, and deciding upon the story elements in it. A third use of narrative is participating in a story, giving the student a role of an actor. Here, the story can be physically experienced by the student and the story is created in a process of living or playing it ‘in real time’. As such, storification becomes a process of acting out the story and the story elements. These three uses of narrative in learning environments can be perceived as various forms of storification, that is, as three ways of experiencing a story. They differ with respect to the amount of activity, with receiving a story as the most passive, and participating as the most active form of storification.

Given the changes in society, and constructivist approaches towards learning, it becomes important that children not only learn by receiving knowledge, but also by searching and creating knowledge. This approach towards learning also means that students should do things that matter to them and to society. Games are argued to be an appropriate means to center the learner, making it possible to learn in a meaningful way, to emphasize problem solving, and to approach learning as an active process of understanding (*Gee, 2005; Prensky, 2003; Shaffer & Gee, 2006; Shaffer, Squire, Halverson, & Gee, 2005*).

In the present study, we evaluated how narrative was designed and applied in a mobile History game in order to get insight into the values of game-based learning as a narrative learning environment in secondary education. We investigated to what extent students received, constructed and participated in the story during the game, as well as what kind of effects these different forms of storification evoked by the pupils.

2. Case study: The mobile History game

The game studied is called ‘Frequentie 1550’, and addresses the history of the city of Amsterdam in late medieval era. It was developed by the Waag Society (<<http://freq1550.waag.org>>), and was piloted in 2005 (see *Admiraal, Raessens, and Zeijts (2007)*). Though it has won a prize of world’s most innovative E-learning application, it has not yet been studied systematically. The mobile game is a one day activity meant for secondary school students to actively experience the history of Amsterdam by walking through the city, experiencing buildings, receiving messages, completing game assignments, while using UMTS/GPS phones for communication and exchanging the information with team members. As has been argued by *Raessens (2006)* this game is a good example of what *Jenkins (2004)* calls ‘spatial games’ that are typically designed as ‘narrative architecture’ or ‘environmental story telling’. Hence, we can expect that this mobile game creates a rich narrative learning environment. We will first explain how the game is structured. More detailed information about the specific content is dealt with in our analysis and will be addressed in the results section.

2.1. The introduction to the game

At the start of the game day, the students gathered at the main location, The Waag of Amsterdam. Here the students were introduced into the main story line of the game, the game structure, the tasks, and the tools to be used.

2.2. Six areas with additional stories

The old city of Amsterdam was divided into six areas, each dealing with a different theme in medieval times. For example, trade is connected to the harbor area of Amsterdam, while defense is connected to the boundaries of the old city. The six themes 'labor', 'trade', 'religion', 'rules and government', 'knowledge', and 'defense' were introduced to the students by means of an introductory video clip that was sent to them as soon as they entered a specific area. These clips presented words that could help the group to complete the assignments of that area.

2.3. Group and team structure

The game was played in groups of four or five students. Each group was divided into a 'city team' (CT) of two or three students who walked through the city and a 'head quarter team' (HQT) of the other two or three group members who stayed behind the computer in the main building. The teams switched places in the second half of the day, so that each student participated in both the HQT and the CT. The CT was assigned one of the six areas as starting point for walking, and was asked to conduct small location-based media tasks in order to explore, map and master the area and its theme, before moving to the learning tasks of the next area. The CT was able to show a map of medieval Amsterdam on their phones. The HQT digitally followed the route of his CT, guided them towards and through the learning tasks by using various information sources. This team also collected all the materials that were produced to complete the tasks. The HQT could use two maps: one about medieval Amsterdam and one about present Amsterdam.

2.4. Three types of assignments in each of the six areas

In each sector there were the same three types of assignments (each composed of three parts): an orientation assignment, an imagination assignment and a symbolic assignment that required some deep learning activities.

The orientation assignment included texts and tasks that intend to trigger environmental awareness by creation or selection of photos, questions about the site and imagination about the viewpoint, work or actions of historical figures. The CT had to act or film (e.g. to carry out work of a shipbuilder, to walk the route of a procession as a pilgrim, or to hang a person) and the HQT had to select those parts of the medieval map that illustrated specific work that was done (e.g. the ships in the harbour that illustrated trade).

The imagination assignment included texts and a task that again intended to trigger environmental awareness by creation or selection of photos and answering questions about the site, but also imagination about historical actions, events or work of historical figures. For this assignment, the CTs were asked to 'act out' particular sayings and tape these on video, and the HQTs were asked to interpret the same sayings. An example is the "fire test", which means "it will be a big test".

The symbolic assignment included texts and tasks for the team to select details of the site that represented a certain symbolic value. The CT was asked to search for and take photos of these details (e.g. "take a picture of the plaque that refers to the medieval name of this area") and the HQT was asked to select the correct picture from various pictures.

2.5. Technical tools

Each CT was provided with a GPS phone to communicate with their group members at the headquarter and a UMTS video phone by which they could receive and send pictures, and text and video messages. Each HQT was provided with a phone, and a computer with an Internet connection and a digital environment to collect assignment material. Each HQT filled a website with all materials that had been collected during the day. All HQTs were provided with two maps: one of the medieval Amsterdam and one of the present Amsterdam, whereas CTs were only provided with the map of medieval Amsterdam. The HQTs could see the routes that all CTs walked on both maps.

2.6. The closing of the game

At the end of the day all students gathered in the main building, where the HQTs were located. Each of the groups was invited to shortly present some of their collected media to the other groups.

3. Methods

3.1. Data

The History game was played by 216 students spread over 10 school classes in five different schools for secondary education. The students ranged in age from 12 to 16 years old; most of them were 13 years old and they varied in educational level from lower secondary professional education to pre-university education. Each school class was randomly divided into groups of four students (and sometimes five), forming two teams of students for each group. In total, 54 groups of students played the game, spread over 10 game days (one school per day). In order to analyze the game design, various materials about and from the game were collected: game descriptions, outline of introductory and

closing presentation, video messages, text messages and assignment texts from the game. In order to examine the storification of the actual game play, the following data were gathered for all groups during the 10 game days:

- a. Observation notes of the researchers throughout the 10 game days about:
 - The introduction to the pupils in the beginning of the game.
 - The involvement of HQTs in the game.
 - The involvement of the coaches of the HQTs.
 - The comments of the pupils at the closing group presentations.
- b. Observations forms completed by the coaches of HQTs and CTs. The observation forms consisted of questions about how much the students collaborated within the team and within the group (HQT and CT), how much they faced technical problems, and how much they were involved in, respectively, the subject, in winning the game, and in solving assignments. For the CT the observation form also included a question about how much the students were paying attention to the historical buildings, and a question about how much problems occurred with respect to walking through the city. For the HQTs it was asked how active they were in searching on the Internet. The questions were answered on a 5-points Likert-scale ranging from 'not' to 'very often'/'very much'. In addition, the observation forms consisted of an open-text box, in which coaches could write all additional matters about the game play they found note-worthy.
- c. Group interview with the coaches at the end of each game day. Topics were similar to the topics of the observation forms of the team coaches. In the group interviews it was checked whether everyone in the group agreed with a particular observation.
- d. Media boxes and logs of the game including:
 - Media boxes of the 54 groups with all exchanged files (text messages, documents, pictures in answer to the assignments) sent by mobile phones from and to the head quarter.
 - Walking path of students in the city.
- e. Observation by the researcher of the group presentation of the students at the end of the day (taped on video).

3.2. Analysis

In order to study the value of the game for learning by storification, the analysis concentrates both on how the game was designed as a narrative learning environment and on how it was actually played as a narrative learning process. With these foci, a content analysis was performed in two steps.

First, we looked at *the design of the mobile History game* and analyzed the way in which the game supports three ways of experiencing a story, respectively, receiving a story, constructing a story and participating in the story. We used the following operational definitions to tag the various game elements with one of the three types of storification:

- Receiving a story: The story is presented to pupils in the form of text, audio or video, in such a way that the pupils can read, listen or watch what is being told, without having influence on the content or the structure of story. Two game elements were identified relating to this category: the introduction of the back story, and the text and video messages sent from the locations of the tasks.
- Constructing a story: The story is written or told by the pupils by interpreting, choosing and organizing story elements, without changing the content of the story elements. Four elements were identified that fit this definition: searching for information on the Internet, answering questions based on collecting information, guiding the CT through the city by HQT, and collecting materials by the HQTs).
- Participating in a story: The story is 'acted out' by the pupils by doing and experiencing the story through the eyes of historical figures, possibly changing the content of the story elements. We identified two game elements that fit this definition: making pictures and videos, and enacting sayings by the CT.

In this way, all game elements were tagged. For each of these tagged game elements the structure and game materials (texts, audio and video fragments, assignments) were described on how each game element was designed and how it represented one of the three types of storification. Three researchers performed this content analysis and negotiated the resulting descriptions until agreement was reached about the details (cf., [Marble, 1997](#)). The resulting descriptions are included in the results section.

Secondly, we looked at *the actual storification process* of the students during the game, by scrutinizing and summarizing the observations of researchers and team coaches, the media boxes and game logs, the group interviews with team coaches, and the final group presentation at the end of the game day, in relation to the nine game elements from step 1. These analyses are summarized in [Table 1](#).

The observation notes from the researchers, the observation forms of the coaches of HQTs and CTs and the group interviews were all analyzed following a grounded theory approach to inductively derive main themes ([Strauss & Corbin, 1998](#)). This led to the separate coding of observations relating to students' involvement (activity and motivation) in competition, collaboration, making assignments, historical content, the game tools, and, for the CT only, for involvement in the city. Besides, codes were identified for observations regarding the technology and regarding the role of the coach. The analysis showed a strong agreement between the coaches. Moreover, the results of the analyses of the observations of the team coaches were in line with the researchers' notes. In order to guarantee the quality of linking the results of the analyses described above to the various game elements, we followed the interrater-agreement procedure described above.

The media boxes and the game logs were used to look how students answered questions and completed their tasks. For example, important for our analysis of the game element 'answering questions' was to what extent the questions were answered correctly and whether this was done in one or more attempts.

These two steps of content analysis (the design and the actual storification process) resulted in three transcripts which were the input for the text reported in the results section, addressing, respectively, receiving a story, constructing a story and participating in a story. These transcripts only include interpretations on which the three researchers reached agreement. In the results section we refer to the source(s) of data on which specific observations are based.

Table 1
Data analyses.

Data	Variables	Game elements	Analyses
Observation notes researchers	Student engagement Student collaboration Learning activities Game activities Role of the coaches	Introduction game day Receiving messages Answering questions Guiding CT Collecting material Walking through the city	Inductively derive main themes Strauss and Corbin (1998)
Observation forms team coaches	Student engagement Student collaboration Learning activities Game activities Technology Their own role	Introduction game day Receiving messages HQ searching for information Answering questions Guiding CT Collecting material Walking through the city	Quantitative data: frequencies Qualitative data: inductively derive main themes
Media boxes and logs	Completion of the tasks	Answering questions Guiding CT Taking pictures and videos Enacting sayings	Frequencies
Group interview with team coaches	Student engagement Student collaboration Learning activities Game activities Technology	Introduction game day Guiding CT Collecting material Walking through the city	Inductively derive main themes
Observation of the group presentation at the end of the day	Student engagement	Enacting sayings	Illustration of other results

4. Results

The results of our game analysis are presented in the form of a narrative on how each of the three forms of storification is embedded in the game design. In addition, we describe the results of the evaluation of how the students applied these elements in terms of attention for the historical content and narrative organization.

4.1. Receiving a Story

Receiving a story is a type of storification that involves narrative organization which is predefined by others, thus triggering reconstruction by the students. The predefined story was contained in the back story of the game, which was partly told in the introduction of the game and elaborated on by video and text messages during the game. There are several aspects that seem significant for evoking storification by students. First, the game required students to watch the narrator and listen carefully to what he told. Second, the storification that was evoked relied strongly on credibility: Whether the pupils were prepared to accept the narrator's perspective, and whether the narrator was able to draw pupils into the world of the story. The credibility in this game is twofold as the story is both fictional (characters are dead, events are not happening now) and real (it describes what really happened). We will now review the game parts that represent a situation of receiving a story, that is, those parts in the game that were predefined and presented to the students by others. We will discuss how these parts were designed and intended, and how students reacted. These results are summarized in [Table 2](#).

4.1.1. Being introduced to the backstory

The game was designed so that before playing the game, pupils were introduced by the game coach to the back story of as follows:

"If you walk out the main door, imagine you walk into medieval times. Therefore, you will also have a medieval map. You are going to try to become Burgher of Amsterdam. Those who stayed in the city for one year and a day were legitimate citizens and became Burgher of the city. Hence, you will collect days instead of points. In the game one can earn 366 days. There is not a real possibility to collect this amount of days, because finishing the complete game probably takes too long. The team with the most days will win the game."

Clearly, this introduction involved some narrative organization, which connects the character of a Burgher to the intention of becoming citizen and to the action of staying 366 days in the city. In this introduction, pupils are implicitly asked to imagine themselves as a Burgher with this intention. To support this imagination, they were provided with a medieval tool: a map of medieval Amsterdam. For this designed back story to succeed, it is important that pupils are drawn into this main storyline of the game.

Table 2
Receiving a story in frequency 1550.

Game elements	As designed	As played
Being introduced to the backstory	Plenary introduction of the back story, procedures, and technology at the Waag	Students paid attention to procedures and technology, instead of story line
Receiving messages from and about medieval time	Video and text messages with assignments, and background information	Students did not pay much attention to the background information and were triggered by completing the assignments

In both the observation reports of the researchers and the group interview with the team coaches it was reported that students paid attention to the presentation of the game and the introduction of the story line, being very silent during the introductory talk. However, it was also reported by both researchers and team coaches that right after the introductory talk, students reacted (with remarks, jokes, and additional questions) on the procedures and the technology of the game, instead of on the story of the game. Evaluation of an earlier version of Frequency 1550 also showed that students did not pay a lot of attention to the back story and that some students mixed the back story of the game and the 'real' historical context (Admiraal et al., 2007).

4.1.2. Receiving messages from and about medieval time

CT and HQT received similar video and text messages about the areas and the themes that are connected to the areas. The introductory video message that started as soon as the CT entered an area, gave an introduction message as if it was being sent from the year 1550. The specific part of the area where the pupils were was shown on the medieval map, and the name of the area appeared in the text, e.g. 'de Lastage'. Then, a series of words appeared that were related to this area (e.g. 'ropes', 'laborers', 'ships', 'defense tower', 'water'), while the sound of the video represented the sound of the area (e.g. sound of manual work, sawing). After these words, the video showed and zoomed in on a few pages from a diary. For the area 'Lastage' the diary reads as follows:

"Tuesday. A few sisters' dresses are worn out and I need a yarn. The best yarn can be found in Lastage because there they make ropes of it. Lastage is a long walk from the monastery, because it lies outside the city wall. As soon as you leave the city wall, you hear the hammering from far away, as in Lastage hard work is conducted. The labourers who work on building and repairing ships always have black pitch stains everywhere. The wind blows hard so the smell of pitch and salty seawater reaches to me. As a woman, I still do not like coming there, but it has not been really unsafe anymore, since the area is being protected by the new defense tower which lies on the outer boundary of Lastage. It is situated at the intersection of the water of the Amstel and the water of the IJ."

Following the diary pages, a closing text appeared (from the main narrator):

"The hidden location is on the external border of Lastage. It is situated at the intersection of the water of the Amstel and the water of the IJ. The hidden location is a tower on the east-side of Lastage."

The video recordings are like micro narratives within the larger narrative of the game that introduced the location area, and the work conducted in this area by particular figures under particular circumstances (in this case, laborers). The diary introduces the perspective of a person in the year 1550, and invites readers to sympathize with the person by describing intentions (e.g. need to buy a yarn), actions and sensed reality (e.g. smells, sense of safety, weather experience). In the video messages, the words, diary text and closing text repeated the main concepts and included signs that point to the hidden location. In the introduction to the game, pupils were advised to pay attention to the video messages as these contained clues that helped to proceed.

When arriving on the assignment locations, the pupils received text messages. Again, these text messages are like micro narrative structures. Consider the following texts for the first assignment:

Part 1:

"In the Middle Ages Lastage was the industrial area of the city. For a long time the laborers worked in dangerous circumstances, as it lied just outside the city wall. In 1516 this tower was built to protect the laborers."

Part 2:

"The laborers in Lastage worked hard. The main part was the shipyard, where ships were built and repaired. The Montelbaanstower built after Lastage was attacked and was burned down by the duke of Gelre."

These texts further introduced the location area, and the work that was conducted by particular characters (laborers) in this area, also pointing to the circumstances. Most of the texts also referred to a specific historical character, (the Duke of Gelre in the example above), and to specific events and outcomes (e.g. attacking and burning down the area, building a tower).

In both the observation reports of the researchers and the observations of the team coaches (as collected in the group interviews and the observation forms) large differences were reported regarding the extent to which students paid attention to the introductory texts and the video messages they received. There were differences between the different days on which different school classes played the game, but also between the different teams on the same game day. One comment made by a researcher that represents an observed difference between days is the following: "There is a clear difference between today and yesterday compared to the game days last week. During these two days students read the texts and watched the video messages much more carefully, rereading them and making notes." Both researchers and team coaches wrote in their observations that generally, students do not read texts carefully. They report how students were scrolling down to the last sentence or part of the message, and then directly moved on to the assignments. For example, in the observation forms one of the coaches noted "the players barely read the background texts".

Both researchers and team coaches reported differences in being involved in the game assignments between groups within a school class. The general tendency not to read the texts is, according to the coaches, because the students were generally more triggered to complete the assignments, than to understand the broader story line that was presented in the received messages. In addition to this explanation, both in the group interviews and on the observation forms the team coaches reported that students had difficulties reading from the videophone (too small and too much sunlight glare).

4.2. Constructing a story

Constructing a story concerns the narrative organization that is intended to be defined by means of students 'telling or writing', be it by extending existing stories or rewriting characters or events into new stories. The challenge of constructing a story for students is that they are required to interpret, choose and organize story elements, such as deciding upon characters and events. Students were invited to be an author of the story in various parts in the game. We can recognize this authoring process primarily in the role and the tasks attributed to

the HQTs, specifically in the four activities of searching and selecting information, answering questions on distance, guiding CT players, and collecting materials. As we will point out in the next sections, some of these activities were the same for the CTs. However, due to the way they were designed, the CT activities related more to the participating in a story than constructing a story. We will review each of these activities of the HQTs, discussing the game design and how the game was played. We summarized these results in Table 3.

4.2.1. Searching for information

Most of the assignments for the HQT included the search for information. This is an author activity in the sense that the information seekers are in control of creating parts of the story and its elements. For several questions in the assignments, the HQT was directly asked to look up information at an encyclopaedia website (such as Wikipedia or historical sites), for example:

“In Wikipedia you’ll find more information about the guilds that were located in the Waag. Find out which guilds belong to which emblems (<http://nl.wikipedia.org/wiki/Waag_%28Amsterdam%29>).”

The quantitative information from the observation forms shows that the HQTs were actively engaged (average score on 5-points scale was 3.52) in surfing on the Internet. However, in some cases, the team coaches explicitly reported differences in the way teams of students searched information on the Internet. In eight forms, the coaches noted that the HQTs could manage to find information quickly, using the websites indicated in the assignment. They reported that “most teams used the right terms for searching the Internet”, “most students knew what key words to use”, and “some teams know how to find more information than necessary, using Internet pages that were not indicated in the assignment”. However, in five forms, the coaches explicitly mentioned that some teams did have difficulty in using the Internet. For example, one team coach mentioned: “Not all teams like working with Wikipedia, because it is unfamiliar to them. In these cases, they started using Google, where these teams often literally inserted the complete question in the search box, instead of using key words. And following, they turned out to be surprised or frustrated when not finding any results”. As such, the search for information only turned out to contribute to constructing a story for those who had some experience in searching information.

4.2.2. Answering questions on distance

Besides looking up information, the HQT had to answer specific questions in different types of assignments. For example, they were asked to select, out of three, the photo that correctly pictured a particular site, or they had to answer a question about the name or meaning of a building, location or historical character, or explain in words a particular historical saying. With respect to storification, the questions they received are intended to trigger awareness of the area in terms of environmental structure (‘the setting’ of the story) and to learn about the work conducted in this area. When looking at the narrative organization involved in these questions, the HQT was invited to link locations to buildings, and working artefacts to intentions and actions. The imagination assignment and the task of explaining a saying in particular, should evoke awareness of historical actions and events, and of historical figures or general characters. For example, the Dutch expression “de vuurproef”, which can be translated literally as “the fire test”, means “it will be a big test.” The meaning of the phrase could be understood once the students discovered the history behind it: an accused had to put his hand in a fire, walk on glowing coals, or walk through a fire in order to prove his innocence. If he was unhurt, he proved to be innocent.

The analysis of the logs of the answers of the HQTs shows that the multiple-choice questions were answered correctly the first time in 75% of the cases; only in 5% of the cases students had to use the maximum of three attempts. The open-text questions, including the explanations of the sayings, required even less attempts of the students before they gave the correct answers: in 81% of the cases student completed the questions correctly in one attempt. If we consider typing errors and grammatical mistakes as incorrect (because the computer did not recognize the answer as correct), this percentage dropped to 60%. The observation reports of the researchers and the quantitative data from the observation forms of the coaches also indicate that the HQTs were actively engaged (average score on 5-points scale was 4.23) in solving the assignments and managed to complete the assignments. This implies that the HQTs understood and expressed the narrative linkages embedded in the questions, as well as the underlying theme of the area.

4.2.3. Guiding CT

Another task of the HQT was to guide the fellow-students from their own CT. This can be considered story construction when they decided upon the routes that their team players should walk to arrive at the assignment locations. The tools they could use were the map of medieval Amsterdam and the modern city map.

Fig. 2 shows the medieval map that was beamed in the head quarter. All HQTs could see how the six CTs (including their own CT) were walking through the city in real time as colored dots. Flags in the map showed where specific assignments were located. Moreover, the medieval map potentially ‘sets the scene’ for storification as it connected these assignments to historical information, and it shows important historical buildings (e.g. defense towers, city walls) and activities (such as the shipping activity at the harbor). Because the HQT was able

Table 3
Constructing a story in frequency 1550.

Game elements	As designed	As played
Searching for information	HQTs should search for information on the Internet	Generally, students used the Internet as expected. There were some exceptions, both positive (searching for more information and in various ways) and negative (not using keywords)
Answering questions on distance	Students were asked to complete assignments and answer detailed questions	In many cases, students answered the question correctly in one attempt
Guiding CT	A map of medieval Amsterdam (for HQTs and CTs) and present Amsterdam (for HQTs only) were available. HQTs could see the route of all CTs	HQTs navigated their fellow-students from the CTs with help of both maps to the assignments. A lot of (phone) communication between the teams was necessary to navigate through the city
Collecting material	HQTs completed assignments and collected information from the CTs in a media box	Students of the HQT were more engaged than students of the CT due to a mix of different tasks and feelings of control



Fig. 2. Map of medieval Amsterdam showing routes of city teams ©Waag society.

to see each player walk through the city, they could work out a strategy and use their phones to guide their city team toward the locations where the assignments were hidden.

As reported by both the researchers (observational notes) and the team coaches (group interviews), the HQTs and the CTs contacted each other regularly, though not always effectively. One difficulty that was mentioned in the group interviews was the timing: sometimes the CT was in the middle of shooting a video, when the phone rang and the pupils had to finish recording. Analysis of the observation forms on both the CTs and the HQTs also shows that there was a lot of communication between the teams about navigating through the city. The question “how much do HQT and CT work together” was mostly (48.3%) answered with “regularly” (4 on a scale of 5). Additionally, 18 team coaches explicitly reported difficulties in communication between HQT and their CT. One of the problems that were mentioned was that the teams are not listening to each other. Some technical problems were also reported, such as dots not appearing or appearing at the wrong place in the map, causing the HQT not being able to detect their fellow-students from their own CT on the maps. Sometimes the team coaches mentioned that the collaboration was not effective as in the following remark “The girls are navigated badly, initially being sent to the wrong tower”. One team coach mentioned that a CT was walking without any help of their HQT: “Navigating was not a problem for the CT. Without help from HQT they walked from assignment to assignment”. Despite the reported difficulties, the observations show that almost all HQTs guided their fellow-students from the CT towards the assignment locations. The HQTs used the medieval map to keep track of their own and other CTs and to have an overview of assignment locations related to a particular area. An example of this use was described by one team coach in the group interview, who mentioned how one girl in the CT quickly ran into a certain direction in order for the HQ to check on the medieval map whether this was showing to be in the good direction towards the assignment location.

4.2.4. Collecting material

The HQTs were not only responsible for answering questions, but also for collecting text, video messages and pictures sent by CTs. This material did not automatically end up in the answering box, but had to be dragged by the HQTs into the so-called ‘media box’ to count as an answer to the questions related to a specific area. The media box can be seen as a potential story board in which the assignment material for one specific area produced by both CT and HQT is collected by the HQTs. As reported by the researchers, the media box was commonly used as ‘home screen’. Together with the medieval map, it appeared to be used as the coordination tool to switch between the different HQ tasks (searching for information, communicate with and direct the CTs, answering questions). In general, the HQTs were very motivated during their play. In the group interviews, the team coaches consistently stated that the students of the HQTs were more busy, motivated, and concentrated throughout game play, compared to their fellow-students of the CTs. This higher motivation of the HQTs seemed to be related to the many different tasks of the HQT, taking the role of director in terms of responsibility and overview of the game play.

4.3. Participating in a story

Participating in a story can be seen as a narrative organization ‘acted out’ by the students. It requires the students to plan, enact, and be aware of narrative elements. Especially the latter could be difficult for students considering the fact that many game elements implicitly carry historical relevant knowledge. An additional difficulty was that the historical facts were partly fictional, and partly authentic (in

terms of ‘accepted as historically true’). Given the above description of participating in a story, this type of storification could be recognized mainly in the activities that were designed for the CTs. Specifically as we will review below, the activities of walking through the city, taking pictures or videos, and enacting sayings, are examples of ‘acting out’ the story. We will describe these designed activities, and discuss how students reacted. These results are summarized in Table 4.

4.3.1. Walking through the city

An important activity for the CTs was walking historically relevant routes through the city. The medieval map was provided, but they did not have a map of modern Amsterdam, like the HQTs did. Also unlike the HQT, the medieval map of the CT showed only one of the six areas of the city and no indications of the assignment locations. This meant that the CT needed the HQT to point out directions.

As described earlier, observation data showed that the CT and the HQT had regular contact, mostly concerning how to walk to the assignments. The CTs had difficulties reading the map of medieval Amsterdam due to the small screen of the video phone and sun glare. This was indicated by both the team coaches and the researchers. Moreover, the medieval map did not represent the current situation in Amsterdam showing a street or water that is not there anymore. Finally, the medieval map (that was used by the students from the CTs) projected Amsterdam upside-down with north being south, and vice versa. As intended in the game design, interdependency in the group was created, as they needed the communication with the HQ teams to be guided through the city. In the group interviews, the team coaches mentioned that CTs got frustrated when they did not receive support, or when they had to wait for it. According to the team coaches the CTs had little overview and a limited sense of the narrative structure of the game, compared to the HQTs. One coach formulated it as follows: “the students of the CT had no clue of what they had done during the game. They did know that they took a picture of the Montelbaanstower, but had no clue why or what for.” The team coaches talked about the lack of ‘flow’ in the gameplay of the CT, due to the long time of walking and waiting between the assignments, as well as some distraction by people, traffic and shops in the street. Accordingly, the historically relevant narrative seems to be lost due to the city experiences of these students.

4.3.2. Taking pictures and videos

In several assignments students were asked to take pictures or shoot videos, some of these requiring a specific angle from which to take these picture and videos. For example, one task was to make a video ‘as if you walk the pilgrim route’. This is a good example of what in the literature has been called ‘perspective taking’ (Järvelä & Häkkinen, 2002). The students are invited to take the point of view of historical characters, evoking imagination by vividly sensing reality from a historical perspective. As to narrative organization, they were implicitly asked to link historical figures to certain actions and to the area.

The ‘perspective taking’ in pictures and the videos varied between the CTs. As pointed out in the group interviews with the coaches, some students tried very hard to make these as consistent with history as possible. In the observation form, one team coach described in an additional note how the CT that he coached asked in a church what nuns wore in that time, so they could try to look like nuns on the video. In some assignments, the perspective taking intended in the game design was not successful, as the students did not understand from what angle they had to take the picture or shoot the video. This was especially the case in the assignment to shoot a video about the area ‘as standing on the Montelbaanstower: 60% of the students recorded the Montelbaanstower itself instead of the area from the perspective of standing on the tower. One possible explanation for this is the earlier mentioned observation that most pupils did not read the tasks very carefully.

Though these tasks (taking pictures and shooting videos) were not always as carefully completed as intended in the design, the team coaches pointed out in the group interviews that students were motivated to complete these assignments. Hence, it seems that they were involved in the historical content embedded in this type of assignments.

4.3.3. Enacting sayings

The imagination task included the task for the CT to act out specific historical sayings that were related to the history of the particular area. This CT task can be considered as a highly active form of storification, as the CT was supposed to act out a historical practice, often in the role of a historical character. The formulation of this task (‘make a video clip in which you act out the saying’) left aside whether the CT was supposed to enact the symbolic or the literal meaning. In the game, both the literal and symbolic expressions in the video were considered to be correct. The CT could ask or receive interpretations of the saying from their HQT.

Almost 100% of the students choose to act out the saying literally. Of these students, 30% of teams choose to combine this with a symbolic enactment or an explanation of the symbolic meaning of the saying. It appeared that the CTs took the freedom the task left them and chose the way to enact that provided most opportunities for acting. For example, the saying ‘the fire test’ was easily acted out literally, but to act out the symbolic value of being a big test was considered much more difficult. With respect to historical storification, both the literal and the symbolic meaning seem valuable and co-reference each other. That is, in the game the sayings all have a symbolic value nowadays, but, at the same time, they point to a historical origin of a certain common practice. On the one hand, the students in the city are faced with this historical practice, often by means of historical tools referred in the sayings. On the other hand, by looking at these historical practice from a contemporary view, they experience the symbolic value. The successful completion of acting out the sayings (both literally and symbolically

Table 4
Participating in a story in frequency 1550.

Game elements	As designed	As played
Walking through the city	Collaboration was necessary to navigate through the city towards the assignment as CTs had only a part of the information needed	A lot of communication between HQTs and CTs about navigation through the city. The students of the CTs experienced a loss of control and overview of the game
Taking pictures and videos	Perspective taking was intended by these assignments	Students were motivated to complete these assignments, although the perspective taking in some assignments was not understood
Enacting sayings	Sayings can be acted out literally, symbolically or both	The video clips showed literal enactments, in one third combined with symbolical meaning of a saying

by one third of the teams) show these students made narrative linkages between characters, their intentions and actions, historically common tools, and the events and outcomes caused by actions.

The pupils were very active and motivated in making these video recordings. This is indicated by the fun and laughter of the pupils seen in the video recordings of the sayings of the media box and seen in the presentations of their video recordings to the other groups at the end of the day.

5. Conclusion

We analyzed a multimedia game supported by mobile technology in terms of how it was designed and how it functioned as a narrative learning environment. Generally, the results showed that many parts of the game evoke narrative organization, which in its most elaborate sense called for the pupils to link characters, tools, and causation by actions and events (see Fig. 1). Our analysis specifically focused on the storification processes induced by the game. The design of the game as well as the actual gaming process was analyzed with respect to how it evoked three types of storification: receiving, constructing and participating in a story. Results show that the game was composed of a mixture of these three types of storification. The analysis revealed how different game elements inferred different ways of how the students relate to the story, in terms of receiving it as spectators, constructing it as directors and participating in it as actors.

The game elements that represented receiving a story were the introduction to the back story presented (face-to-face) in the morning to all the students, and the text and video messages sent to them at the assignment locations. The game elements that represented constructing a story were found to be related to the role of the HQTs, who collected, coordinated and directed the game. Specifically, the construction process took place during four activities: searching and selecting information, answering questions on distance, guiding CT players, and collecting materials. The participating storification processes induced by the game were found to be related to the role of the CTs who completed location-based assignments of taking pictures, recording videos and enacting sayings.

Comparing these three storification processes, we can see that both constructing and participating are processes in which the learner is the author of the story in the sense of having influence on if, when and how narrative linkages are made. This can be argued much in line with a constructivist approach to learning. In case of participating, the activity can be considered more intense than in constructing, as the story was being literally acted out and 'lived', and buildings, viewpoints and activities of historical characters were physically perceived and, therefore, more deeply experienced. However, based on the findings, we can say that in case of participating, the focus of the pupil was also distracted by all that is happening in real time in the street. While trying to find their way through the city, searching for assignment locations and completing the assignments, students lost the sight of the overall structure of the game and its narrative. As Aylett pointed out in relation to experience of participation (p. 9): "Participation is also a totally different experience from spectating, since the commitment to act (having to decide what to do next) and the necessarily partial perspective of a specific character radically changes the input to the storification process". Apparently, the bird perspective that one has in case of constructing a story, is precisely what is lost in processes of participation. When the HQTs switched between various sources (Internet, CT, media box and the medieval map) and tasks (answering questions, searching for information, guiding the CT, collecting answers), a broader picture was created. In contrast, for the HQT it remained more abstract about what building, historical saying, viewpoint, character the other students were talking. By physically experiencing these narrative elements, they become more meaningful for the CT.

Though storification is enhanced by 'living the narrative', as in case of participating, it is also central to storification to have narrative overview as in case of constructing. This raises the question how the various ways in which storification can take place can be applied into games in such a way that one creates enough *involvement in the whole story*, while, at the same time, maintaining enough *meaningful experience of the narrative elements*. In this game, this was successfully combined, by the creation of two different roles, to be played by all students. Future research on games and game design should explore whether it is possible to further merge these storification processes. The perspectives of the three distinguished storification processes allow us to look more deeply into sense-making processes in such narrative structures as games.

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